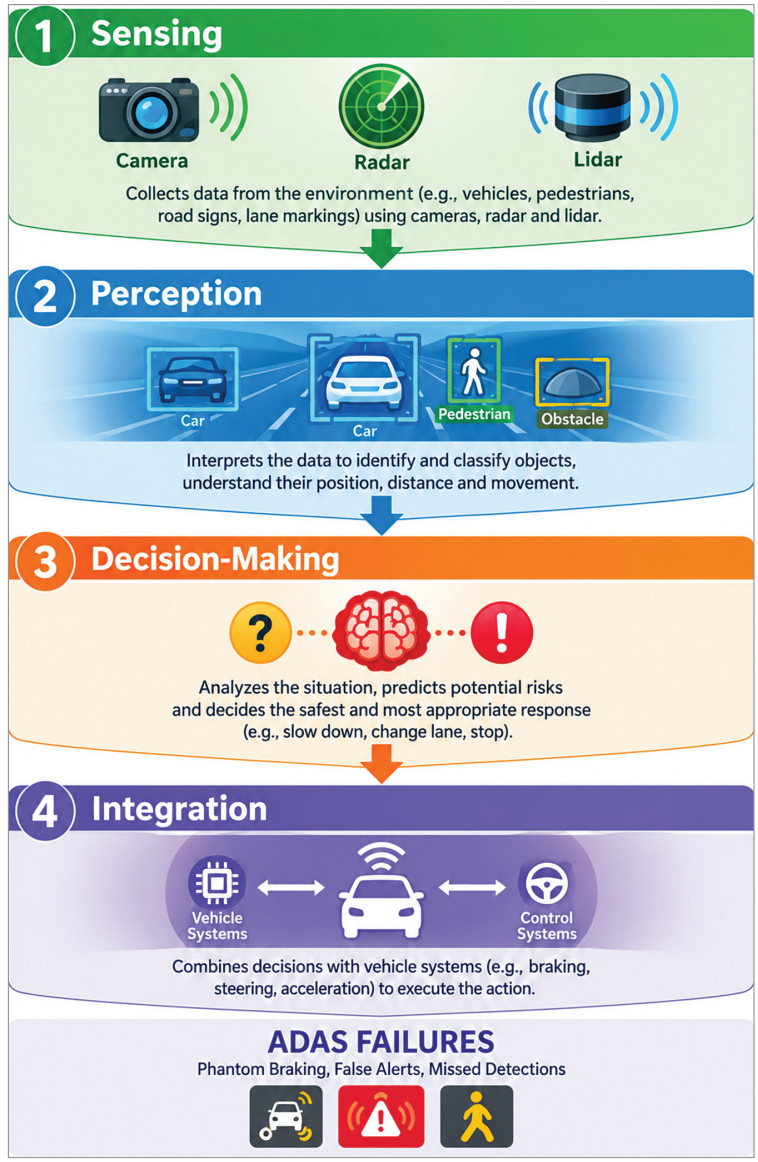
affect a camera's field of view.

The road environment itself introduces additional complexity. The world is not always well-structured or easy to interpret. Lane markings are often ambiguous, serving more as suggestions than clear definitions. Traffic comprises pedestrians, bicycles, two-wheelers, and many other types of vehicles, each moving according to its own often-unwritten behavioural patterns.

So what happens when a system trained on structured scenarios, is presented with this kind of ambiguity? The problem becomes even more pronounced when movement is introduced. A pedestrian changes direction mid-step. A vehicle swerves unexpectedly. An object appears without prior context. These are situations that must be interpreted in real time. Many of these scenarios are also underrepresented in training datasets.

There is another, more subtle challenge: Degradation rather than outright failure. Dust on camera lenses and dirt on sensors can gradually impair performance. Partial occlusion may not be sufficient to trigger a system alert. The system may continue functioning normally while operating with a diminished view of the environment, often without informing the driver.

This raises a sharper question: Are these systems engineered for the real world or for a simplified representation of it? Technology is not proven under ideal conditions.



It is proven amid noise, unpredictability, and constant deviation from the expected. That is where resilience is tested. Increasingly, that is where the underlying fragility of many ADAS architectures begins to surface, not as catastrophic failure, but as inconsistency at precisely the moments when consistency matters most.

Understanding the root causes of system failures

To understand the challenge, it is important to recognise that ADAS failures rarely result from a single point of failure. They are systemic, layered, and often unfold as a chain reaction.

The first layer is hardware sensing. Cameras, radar, and lidar each provide distinct advantages while carrying inherent limitations. Cameras struggle with glare and poor visibility. Radar can misinterpret reflections. Lidar, while highly precise, is not immune to environmental noise. These are not failures in isolation; they are constraints. However, when distorted inputs enter the system, they influence everything that follows.

The next layer is perception, the system's version of 'seeing.' Under ideal conditions, perception is remarkably precise. However, the real world does not always present clean inputs. A shadow stretching across the road may mimic an obstacle. An object may appear only partially, obscured or outside the camera's field of view. In these situations, perception becomes probabilistic

From insight to action: Designing for what matters most

To move forward, ADAS teams should align around a few core principles:

- Design for edge cases, not just averages
- Build sensor stacks that are complementary, not redundant
- Prioritise confidence and consistency over isolated accuracy gains
- Continuously validate against real-world variability

Within this framework, the role of thermal imaging becomes clear. It is not about replacing existing sensors but about strengthening the system where it is most vulnerable.